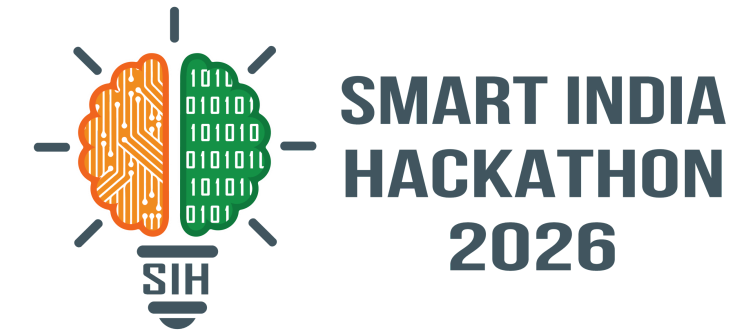


SMART INDIA HACKATHON 2026



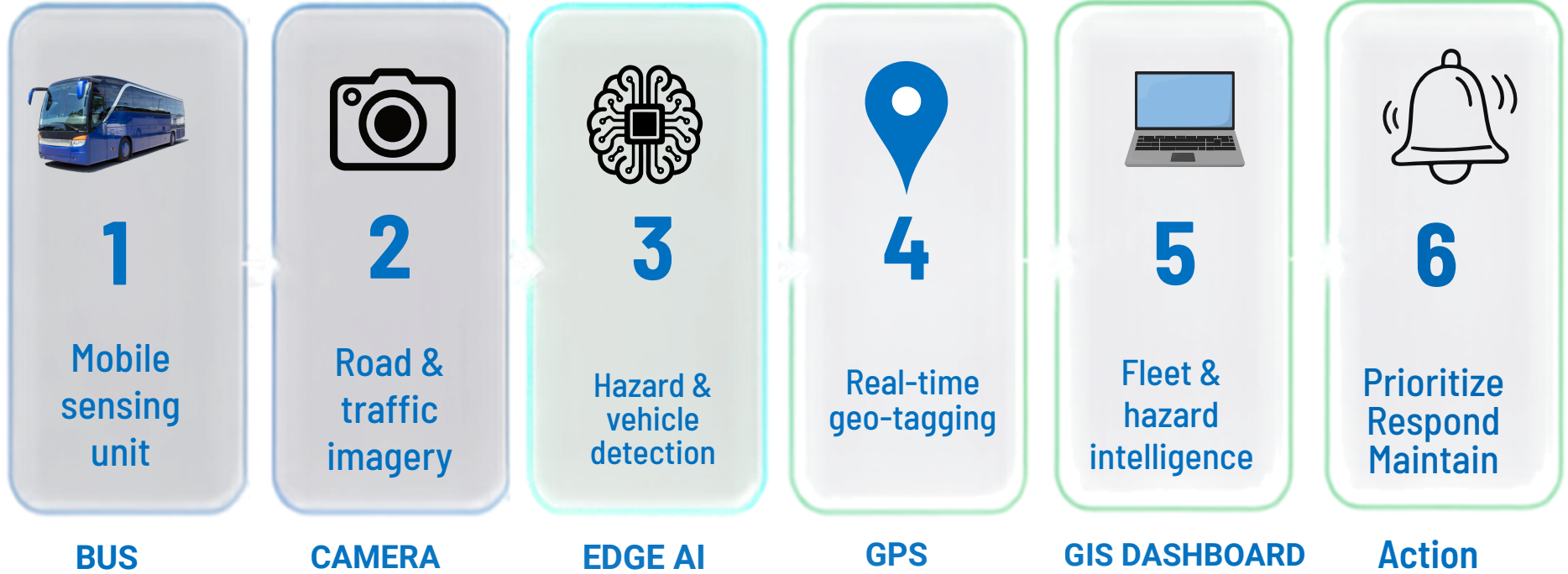
- Problem Statement ID - **SIH26124**
- Problem Statement Title - **AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet**
- Theme - **Smart Automation**
- PS Category - **Software**
- Team ID - **SIH2026048**
- Team Name - **TransitMind**



CHALLENGES

-  **Limited Fixed Coverage**
Fixed CCTV cannot continuously monitor every road segment.
-  **Manual Inspection**
Periodic inspections are time-consuming.
-  **Reactive Complaints**
Citizen reports often identify issues after they become visible problems.
-  **Delayed Response**
Authorities may lack timely location-specific information.

PROPOSED SOLUTION



HOW IT ADDRESSES THE PROBLEM

Continuous Sensing → AI-based Detection → GPS Verification → Centralized GIS View → Prioritized Response
Turns delayed, manual road monitoring into continuous, location-aware urban intelligence.

EDGE AI

AI inference runs locally on the bus/smartphone edge unit.

LOW BANDWIDTH

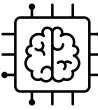
Only lightweight telemetry and verified event snapshots are transmitted.

FLEET SCALE

Multiple buses continuously collect road-condition information across routes.

Turn everyday public buses into a city-wide, continuous urban sensing network.

INNOVATION HIGHLIGHTS

-  **Mobile Sensing Network**
Uses existing public transport movement as a sensing network.
-  **AI + GPS Intelligence**
Combines computer vision with location information.
-  **Multi-Bus Coverage**
Multiple buses continuously collect road-condition information.
-  **Spatial-Temporal Intelligence**
Groups nearby/repeated detections to reduce duplicate reports.

1 EDGE SENSING

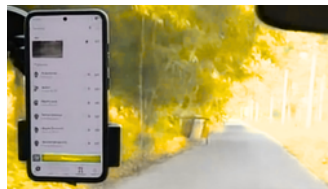
Bus Fleet / Smartphone



Public bus cameras



**SMARTPHONE
EDGE NODE**



**Smartphone
camera + GPS**

2 DATA INGESTION

WebSockets → FastAPI
Backend



Camera frames + GPS
telemetry → backend



Real-time Data stream
JPEG Frames + WebSocket

3 AI DETECTION

Camera Frames



MULTI-TASK DETECTION



Water
Waterlogging



Safety
School-Zone
Safety



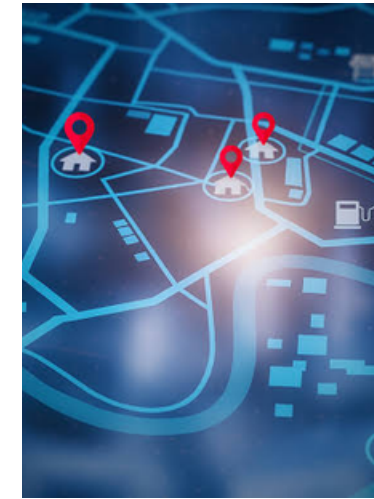
Traffic
Vehicle
Counting



Hazard
Road
Hazards

4 LOCATION INTELLIGENCE

Multiple observations → One record



Latitude



Longitude



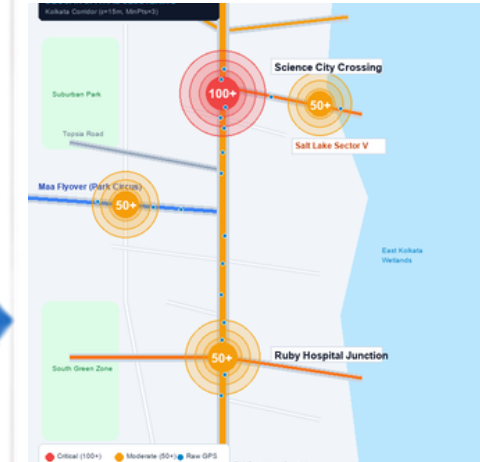
Timestamp



Route Context

5 DETECTION CLUSTERING

Camera Mapping → Others



**Multiple bus
observations → One
record**



**Reduce duplicate
reports**

6 URBANSENSE GIS DASHBOARD



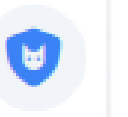
MUNICIPAL AUTHORITY



**Road
Maintenance**

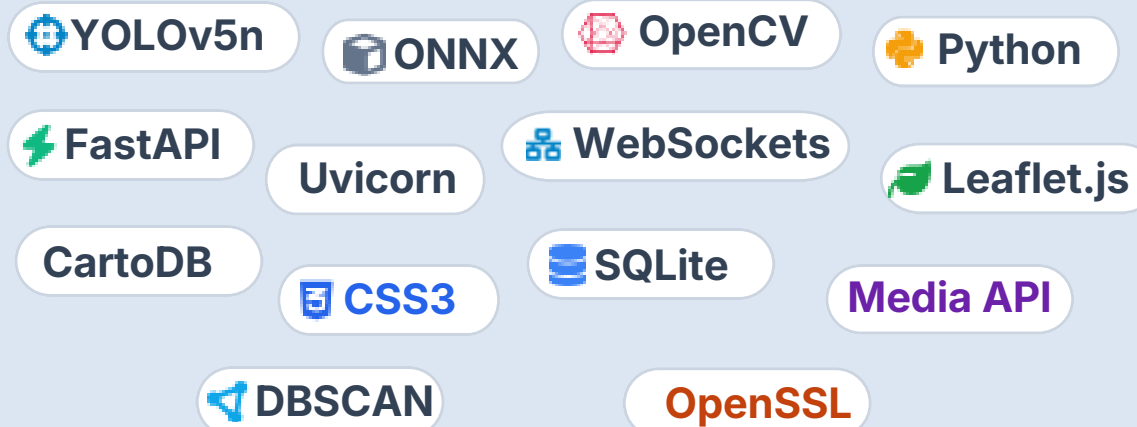


**Traffic
Management**



**Public
Safety**

TECH STACK



EDGE PROCESSING

- Local AI inference
- Reduced dependence on raw-video transmission

REAL-TIME SENSING

- Camera + GPS
- Live event/telemetry flow

FALSE-ALARM CONTROL

- Surface + cavity verification
- Verified hazard-hazard events

FLEET INTELLIGENCE

- Multiple bus observations
- Spatial-temporal deduplication

Designed for practical deployment using existing fleet infrastructure

FEASIBILITY



TECHNICAL

- YOLOv5n ONNX edge inference
- Real-time lightweight telemetry
- Modular FastAPI/WebSocket architecture



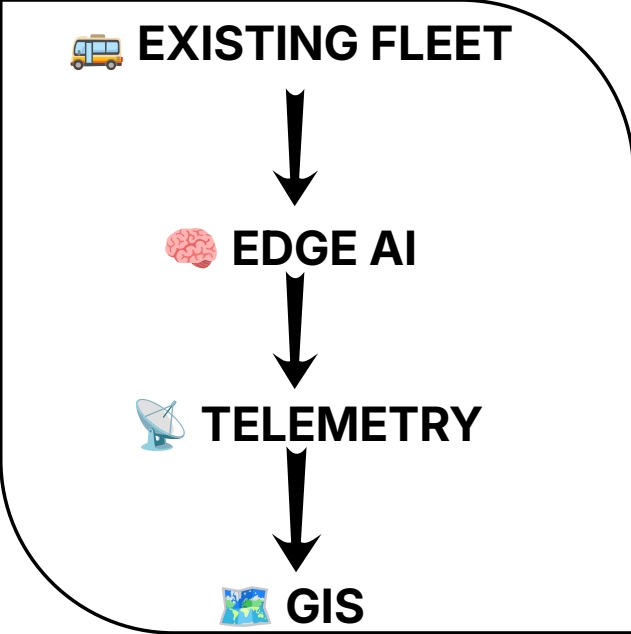
OPERATIONAL

- existing public transport routes
- Smartphone & edge AI nodes on fleet
- Centralized municipal GIS monitoring



ECONOMIC

- Reuses existing fleet infrastructure
- Low-cost edge nodes
- Zero external cloud dependency



CHALLENGES



Network interruptions



GPS drift & urban canyons



Lighting glare & shadows



Multi- bus report overlap



Fleet-scale concurrency



Camera vibration & blur

MITIGATION



Connectivity Loss → offline_queue



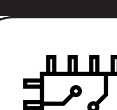
GPS Error → spatial-temporal clustering



False Detection → pavement texture + cavity verification



Duplicate Reports → DBSCAN clustering (15m / 60s)

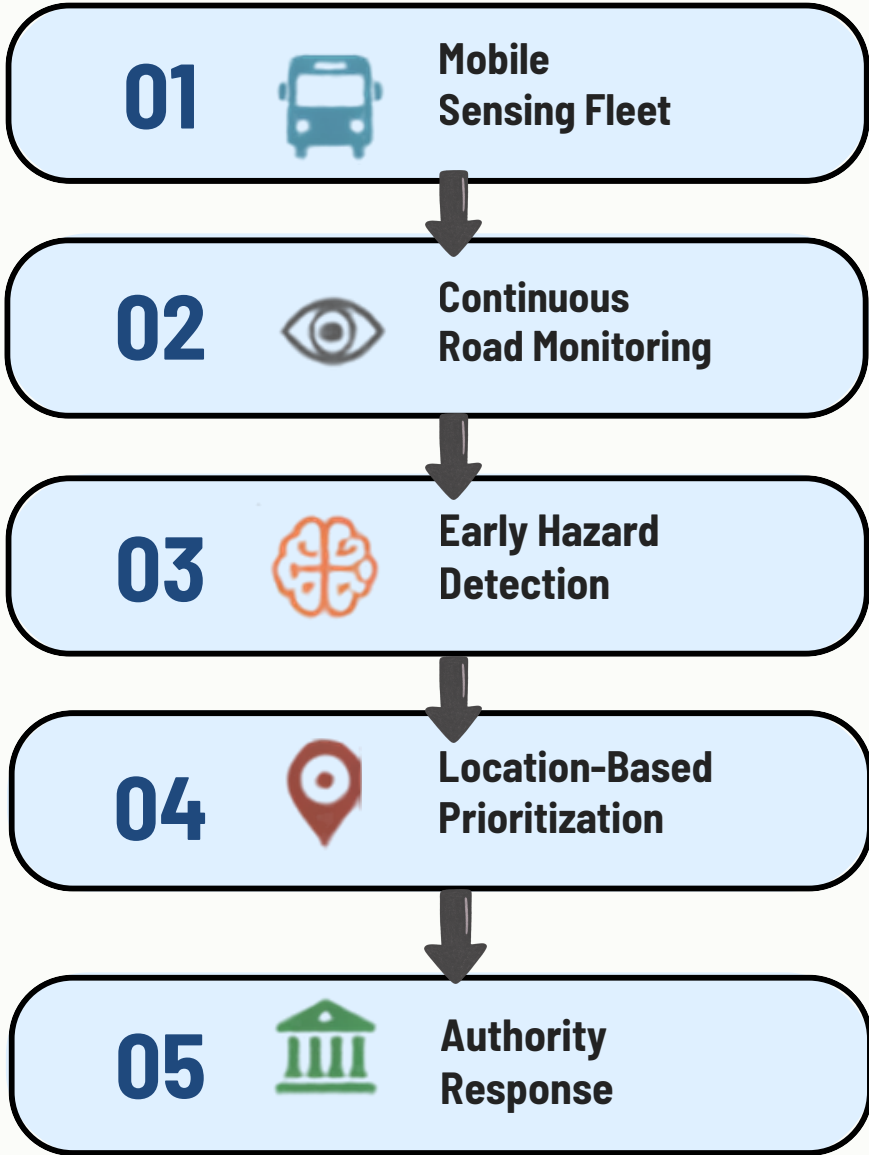


Fleet Scaling → modular FastAPI/WebSocket architecture

PROTOTYPE → MULTI-BUS DEMO → FLEET SCALE → CITY-WIDE SCALABILITY

Current prototype demonstrates the end-to-end workflow; large-scale deployment requires field validation and infrastructure scaling.

IMPACT AND BENEFITS



Social Impact

- Safer mobility
- earlier hazard identification
- better road condition awareness
- improved public transport
- reliability

Economic Impact

- smarter resource allocation
- less manual survey dependency
- prioritized maintenance
- preventive planning

Environmental Impact

- more efficient mobility
- better route planning
- fewer inspection trips
- traffic-aware routing

Governance Impact

- evidence-based decisions
- centralized intelligence
- location-aware maintenance info
- historical hazard DB
- scalable corridor monitoring

FROM PERIODIC INSPECTION → CONTINUOUS INTELLIGENCE

TRADITIONAL APPROACH

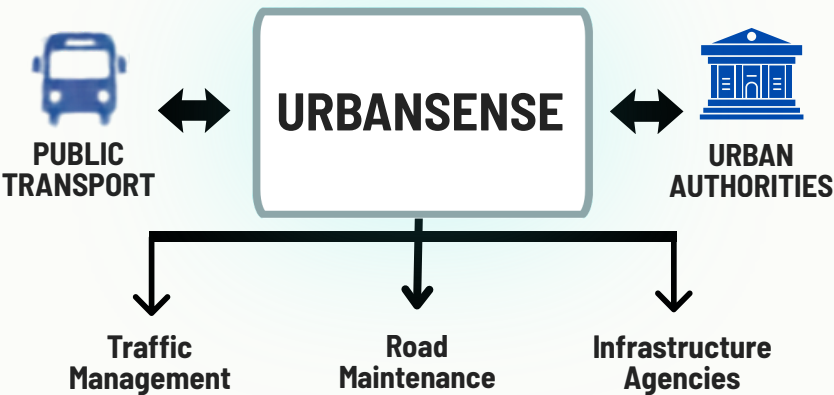
- Manual audits
- Citizen complaints
- Fixed CCTV

Delayed / incomplete awareness

BEL URBANSENSE

- Moving bus fleet
- Edge AI
- GPS + GIS
- Verified events

Continuous urban intelligence



WHO BENEFITS?

MUNICIPAL / PWD

Road maintenance and work-order prioritization

TRANSPORT AUTH.

Fleet monitoring and route management

TRAFFIC POLICE

Congestion and incident awareness

CITIZENS

Safer and better monitored road

TRANSIT OPERATORS

Hazard-aware fleet operations

01 | GOVERNMENT & STANDARDS

MoRTH • Road Accidents in India

Road safety and accident statistics
morth.nic.in

AIS-140 • Vehicle Tracking Standard

GPS tracking & emergency communication
<https://morth.gov.in/sites/default/files/AIS-140.pdf>

Indian Roads Congress

IRC:SP:84-2019 / IRC:SP:87-2019

Road Infrastructure Standards
Reference: irc.nic.in

02 | TECHNICAL & RESEARCH

Ultralytics YOLOv5

Edge object detection
github.com/ultralytics/yolov5

DBSCAN • Ester et al., KDD 1996

Spatial clustering & deduplication
<https://cdn.aaai.org/KDD/1996/KDD96-037.pdf>

OpenCV DNN

ONNX / neural network inference
docs.opencv.org/.../dnn.html

Leaflet.js

Interactive GIS visualization
<https://leafletjs.com/>

03 | WORKING PROTOTYPE & SOURCE

BEL UrbanSense Dashboard

<https://bel-urbansense.onrender.com>

Mobile Edge Camera

<https://bel-urbansense.onrender.com/camera>

FastAPI / API Docs

<https://bel-urbansense.onrender.com/docs>

GitHub Source Code

https://github.com/SourishGhosh-26/pi_pothole_detection

RESEARCH BASIS

 Edge AI •  Road-Surface Perception •  GPS Telemetry •  Spatial Clustering •  GIS Visualization